

Composition Pair A1.01 + B1.20: Human Governance Provides the Core Governance Principle That Recursive Governance Extends to Every Structural Level, Making the CKS Architecture Coherently Human-Governed Throughout

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 13, 2026

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026). It does not introduce new axioms. Its contribution is to formalize, in operational form, the composition relationship between two architectural commitments — human governance (A1.01) and recursive governance (B1.20) — and to establish that the two together constitute the CKS architecture’s complete governed architecture claim.

Abstract

Among the composition pairs in the CKS derivation series, A1.01 + B1.20 carries the greatest foundational weight: it is the pair in which Paper 1’s core governance commitment and Paper 2’s most significant governance extension are formalized as a unit. A1.01 (human governance) is the governance principle for the CKS architecture: humans govern AI behavior through the governed substrate, with governance exercised via orchestration rules and governance affordances that preserve human rights to inspect, modify, and override at any time. B1.20 (recursive governance) is the architectural extension of that principle to every structural level Paper 2 introduces: the same governance commitment that A1.01 establishes at deployment scope applies, by recursive inheritance, at cell scope, aspect scope, and Self scope independently. Without A1.01 there is no governance principle for B1.20 to extend; without B1.20, A1.01 applies only at deployment scope, leaving individual entities ungoverned at their own level. Together, the pair closes every entity-level governance gap and constitutes the architecture’s claim to be coherently human-governed throughout — not merely nominally governed at the container level.

1. Why this pair has foundational architectural weight

The CKS derivation series documents composition pairs: two architectural commitments that require each other to function as claimed, each enabling and depending on the other in ways that neither

commitment alone can substitute for. Most composition pairs relate two commitments at the same architectural level or within the same phase of the theory. This pair is different: it relates Paper 1's foundational governance principle to Paper 2's most significant governance contribution. It is the pair in which the governance architecture's evolution from Paper 1 to Paper 2 is made explicit.

Paper 1 established A1.01 at single-cell scope: humans govern AI behavior through the governed substrate; the three governance rights (inspect, modify, override) are available to humans at all times over substrate content and orchestration rules; governance is an authority architecture, not a review workflow. This commitment makes CKS a governed architecture rather than merely a functional one — it is what separates a deployment that happens to work from a deployment that is structurally accountable to human authority.

Paper 2 introduced three structural levels (cell, aspect, Self) and with them a question Paper 1 had no occasion to address: does A1.01's governance principle apply at the level of individual entities — cells, aspects, Selves — or only at the deployment container? B1.20 answers this question: Paper 1's commitments, including A1.01, apply recursively at every structural level. Governance is not a deployment-level property that individual entities inherit passively; it is an entity-level property that each structural scope independently satisfies.

This makes B1.20 the most significant governance contribution in Paper 2. It is not an incremental extension; it is an architectural generalization that changes the scope of the entire governance claim. And it makes the A1.01 + B1.20 pair the governance architecture's completeness argument: A1.01 supplies the governance content; B1.20 supplies the governance scope.

2. A1.01 as the governance principle: who, how, and what

A1.01 establishes human governance as one of the CKS architecture's foundational commitments. It can be stated along three axes.

Who governs. Humans govern. Not the LLM, not runtime middleware, not the substrate itself in any autonomous sense. Governance authority is held by the humans who can exercise the three rights that constitute it: the right to inspect any substrate content and orchestration rule at any time; the right to modify any substrate content and orchestration rule; the right to override any operation, default, or AI-produced output that touches substrate content or orchestration rules. These rights belong to humans architecturally — as a property of the system's design — not as a procedural promise contingent on any particular deployment workflow.

How governance is exercised. Governance is exercised through the substrate. Orchestration rules, authored by humans and recorded in the substrate as DNA-layer content, determine how cells behave. The substrate mediates all AI activity — AI reads from and writes to the substrate under those rules; the rules define what the AI is authorized to do; humans author the rules and retain the right to revise them at any time. Governance does not require per-element human review of substrate content; it requires that the authority structure is always in place and always exercisable. The two moments at which governance is exercised — orchestration rule authoring (design time) and direct override

(intervention time) — are neither proportional to substrate size nor dependent on human performance of routine labor.

What is governed. AI behavior is governed — specifically, AI behavior via the instinct layer. The substrate governs what the LLM is authorized to do on the substrate’s behalf, how its outputs are integrated into substrate state, and what conditions trigger what responses. Governance of AI behavior is not governance of LLM weights or inference; it is governance of the coordination context in which the LLM operates and of the rules that determine how its outputs become substrate state.

A1.01 is the commitment that makes CKS a governed architecture. A deployment that satisfies the other CKS commitments — substrate-as-coordination-artifact, conflict preservation, tool-agnosticism, linear-cost scaling, path retraceability — but not A1.01 may be a functional architecture; it is not a governed one. A1.01 is the foundational commitment without which the governance vocabulary the architecture uses has no anchor.

3. B1.20 as the recursive extension: what it adds and what it requires

B1.20 extends A1.01’s scope from deployment to entity. It does so by establishing a recursive commitment: Paper 1’s architectural commitments — all of them, with A1.01 among them — apply at every composition level of the three-level architecture Paper 2 introduces.

Without B1.20, the governance situation in a three-level CKS deployment is as follows. The deployment as a whole satisfies A1.01: there is a substrate, orchestration rules are human-authored, the three governance rights are available. But individual cells operate under those rules without independently satisfying A1.01 at their own level; individual aspects coordinate cells without independently satisfying A1.01 at the aspect level; the Self integrates aspects without the integration logic independently satisfying A1.01 at the Self level. Governance applies at the container but not coherently at the contents. Individual entities may exhibit AI behavior that is not governed at the entity level — that is authorized only by the deployment-scope rule set, without level-appropriate human governance at the entity’s own structural scope.

With B1.20, this situation is closed. A1.01 applies at cell scope: each cell independently satisfies human governance of AI behavior through its own DNA layer — orchestration substrates, behavior substrates, schemas — authored and governed by humans at the cell level. A1.01 applies at aspect scope: each aspect independently satisfies human governance of the coordination it performs across its constituent cells. A1.01 applies at Self scope: the Self independently satisfies human governance of the integration it performs across aspects. No entity at any structural level is ungoverned at its own level.

B1.20 makes this recursive application an architectural commitment, not a deployment policy. The commitment is structural: the same governance-boundary inheritance that Paper 1 §4 establishes for the cell holds at every composition level Paper 2 adds.

4. Mutual dependency: each requires the other

The pair is mutually dependent in a strong sense. Neither commitment is architecturally complete without the other.

A1.01 requires B1.20 for architectural completeness. A1.01's governance principle — humans govern AI behavior — is fully realized only when governance applies at the level where AI behavior actually occurs. AI behavior in a three-level CKS deployment occurs at the entity level: cell instinct operations, aspect coordination operations, Self integration operations. If A1.01 applies only at deployment scope, there are entity-level gaps. A cell performing instinct operations without level-appropriate human governance is exhibiting ungoverned AI behavior, even if the deployment container is nominally governed. B1.20 is what closes those gaps by making A1.01 apply at every entity scope.

B1.20 requires A1.01 as its governance content. B1.20 is a recursive application commitment: it specifies that Paper 1's commitments apply at every level. But what B1.20 applies recursively must be something. A1.01 is the primary governance commitment being extended. Without A1.01, B1.20 would be a recursive architecture — a framework for applying commitments across structural levels — without a foundational governance commitment to extend. The recursive architecture is valuable precisely because it extends a principle that matters: human governance of AI behavior. B1.20's architectural significance is that it extends A1.01's governance content to every entity at every level. Without that content, B1.20 recurses a void.

The mutual dependency has a practical consequence: implementations that satisfy one commitment without the other produce either an empty recursion (B1.20 without A1.01) or a deployment-level-only governance shell (A1.01 without B1.20). Only the pair together produces the result the CKS architecture claims.

5. Co-presence requirement: what each absent state produces

The three states — A1.01 without B1.20, B1.20 without A1.01, and full co-presence — are architecturally distinct and produce different deployment conditions.

A1.01 without B1.20 — governance principle without recursive extension. The deployment satisfies A1.01 at deployment scope. Orchestration rules are human-authored; the three governance rights are available; the substrate governs AI behavior at the container level. But individual entities operate without level-appropriate human governance at their own structural scope. A cell's DNA layer may not independently satisfy A1.01 at cell scope; an aspect's coordination logic may not independently satisfy A1.01 at aspect scope. The deployment appears governed; individual entities are not governed at their own level. This is the Deployment-Level-Only Governance failure mode (B3.21): governance present in form at the container level, absent in substance at the entity level.

B1.20 without A1.01 — recursive architecture without governance principle. The architecture commits to recursive application of Paper 1's commitments at every level, but the primary commitment being applied — human governance of AI behavior — is not well-defined. B1.20 provides the application machinery; A1.01 provides the substance to be applied. Without A1.01, B1.20's recursive

commitments are structurally present but substantively empty. This is governance theater of a structural kind: the recursive architecture exists but does not carry a meaningful governance commitment at any level. The Governance Theater failure mode (B3.22) names the condition in which governance is present in form but not in substance.

Full co-presence — A1.01 + B1.20 operational together. A1.01 governance applies recursively at cell, aspect, and Self scope per B1.20. Each entity independently satisfies human governance of AI behavior at its structural scope. Every cell can demonstrate that its DNA layer is human-governed, that its orchestration substrates are human-authored, and that the three governance rights are available over its substrate content. Every aspect can demonstrate the same at aspect scope. Every Self can demonstrate the same at Self scope. The deployment is coherently human-governed throughout — not merely at the container, but at every entity at every level.

6. The Paper 1 → Paper 2 architectural progression

The pair A1.01 + B1.20 represents the governance architecture's evolution from Paper 1 to Paper 2, and this progression is worth naming explicitly as architectural prior art.

Paper 1 established A1.01 at single-cell scope. At that scope, the governance question is simple: the deployment has one structural level, and A1.01's governance principle applies at that level. There is no multi-level architecture to worry about; the governance scope is the deployment scope.

Paper 2 introduced three structural levels. With three levels comes the question that Paper 1 had no occasion to address: does governance apply at the level of individual entities, or only at the deployment container? A deployment with three structural levels where governance applies only at the container has two structural levels — aspect and Self — where AI behavior may be ungoverned at the entity's own scope.

B1.20 is Paper 2's answer: governance applies at every structural level. This is the most significant governance contribution of Paper 2. It is not a restatement of Paper 1; it is a generalization that the multi-level architecture makes necessary. Paper 1's A1.01 at single-cell scope was architecturally complete for single-cell deployments. B1.20 makes A1.01 architecturally complete for multi-level deployments by extending the governance scope from deployment to entity at every level.

The pair together documents this progression as a publicly established prior art claim: the CKS architecture's governance completeness argument has two components — the governance principle (A1.01, Paper 1) and its recursive extension to multi-level scope (B1.20, Paper 2) — and neither component alone suffices for the complete claim.

7. Operational integration and detection

Operational integration. In a deployment satisfying both commitments, the A1.01 governance affordances are exercised at both deployment scope and, per B1.20, at entity scope. At cell scope, human governance per B1.20's recursive A1.01 is implemented through the cell's DNA layer:

orchestration substrates, behavior substrates, and schemas authored and governed by humans with the three governance rights preserved over them. At aspect scope, governance applies to the coordination logic that operates across constituent cells. At Self scope, governance applies to the integration logic that holds aspects as facets of one CKS-governed intelligence. The recursive commitments are verified at each scope independently.

Detection. The pair is operationally present when the following conditions hold: (B2.102 — recursive A1.01) Is A1.01 applied at each entity level? Can each cell, aspect, and Self independently demonstrate that its substrate content and orchestration rules are human-governed, that the three governance rights are available, and that governance is an architectural property rather than a procedural promise? (B2.110 — recursive commitments verification) Is the complete pair operational throughout the deployment? Can governance for any entity at any level be traced to level-appropriate A1.01 governance per B1.20?

Compliance demonstration. The pair together enables entity-level compliance demonstration — the practical governance advantage that distinguishes the A1.01 + B1.20 architecture from deployment-scope-only governance. A question such as “does this cell satisfy human governance?” receives a determinate answer because B1.20 specifies that A1.01 applies at cell scope, and the cell’s DNA layer either satisfies the A1.01 criteria or it does not. Without B1.20, this question is not answerable at cell scope — governance is only demonstrable at deployment scope — and the entity-level compliance claim cannot be made.

8. Operational test

A deployment instantiates the A1.01 + B1.20 composition pair if and only if all of the following hold:

1. A1.01 is satisfied at deployment scope: humans retain the rights to inspect, modify, and override substrate content and orchestration rules at any time; those rights are architectural, not procedural.
2. A1.01 is satisfied at cell scope: each cell’s DNA layer — its orchestration substrates, behavior substrates, and schemas — is human-governed, with the three governance rights available over cell-level substrate content independently.
3. A1.01 is satisfied at aspect scope: each aspect’s coordination logic is human-governed, with the three governance rights available over aspect-level substrate content independently.
4. A1.01 is satisfied at Self scope: the Self’s integration logic is human-governed, with the three governance rights available over Self-level substrate content independently.
5. The recursive application of A1.01 across levels is an architectural commitment of the deployment, not a deployment policy or operational practice that may vary.
6. Governance at each entity scope is demonstrable independently: an auditor examining any entity at any level can confirm A1.01 compliance for that entity without reference to the deployment container’s governance state.

A deployment that satisfies (1) but fails (2)–(4) instantiates A1.01 without B1.20 — Deployment-Level-Only Governance (B3.21). A deployment that satisfies the structural recursion of (2)–(5) without a coherent governance principle at (1) instantiates B1.20 without A1.01 — Governance Theater (B3.22). Only the deployment satisfying all six conditions instantiates the complete pair and can make the CKS architecture’s coherent human-governed claim throughout.

9. Conclusion

A1.01 + B1.20 is the governance architecture’s foundational pair: the governance principle and its recursive extension to every structural scope. A1.01 establishes that humans govern AI behavior through the governed substrate, specifying who governs, how governance is exercised, and what is governed. B1.20 establishes that this governance principle applies not only at deployment scope but at every entity scope — cell, aspect, and Self — in the three-level architecture Paper 2 introduces. Neither commitment is architecturally complete without the other: A1.01 without B1.20 leaves entity-level governance gaps that deployment-scope governance cannot close; B1.20 without A1.01 recurses a governance commitment that is not defined.

Together, A1.01 + B1.20 constitute the CKS architecture’s complete governed architecture claim: human governance of AI behavior at every structural scope throughout the deployment. This is the governance architecture’s evolution from Paper 1 — which established A1.01 for single-cell deployments — to Paper 2 — which extends A1.01 to the multi-level architecture B1.20 governs coherently throughout.

Subsequent work that adopts, extends, or argues against the CKS governance architecture should treat this pair as a unit. A deployment that satisfies A1.01 at deployment scope and B1.20’s recursive architecture but not their conjunction has neither.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

How to cite this note

Li, W. (2026). *Composition Pair A1.01 + B1.20: Human Governance Provides the Core Governance Principle That Recursive Governance Extends to Every Structural Level, Making the CKS Architecture Coherently Human-Governed Throughout*. May 13, 2026. ORCID: 0009-0004-8065-3235.